Introduction to the Mathematics of Finance. Take-Home Midterm.

Due October 19, 2026 11.59 p.m.

Please write a pledge that you did not copy solutions from the work of other students. You can consult TAs if you have any difficulties. This midterm uses content of lectures in the last two weeks of February. It also uses base matlab models that can be downloaded from Courseworks.

1. On Mac computer start Terminal go to shell, select New window. New terminal appears. Type at prompt.
`jupyter notebook` jupyter opens in a browser.
In jupyter in browser: Go to File, Open, open file "BrownianMotion.ipynb" that you previously downloaded from coursework.
Run notebook (Shift-Return).
Modify that norebook to Geometric Brownian motion with the starting value Xo=100, growth rate $\mu = 0.10$, volatility $\sigma = 0.21$ and 5000 trajectories.
Check that the code works. Try out 50,000 trajectories. Try out 100,000 trajectories. Submit the code printout and the graph printout for 5000 trajectories.

2.Python option model. Download from courseworks python option model file "BlackScholesStocks.ipynb".
Open python notebook.
Run it and the graph will appear.
Now modify the file BlackScholesStocks.ipynb so that the function now calculates the price of options on stocks paying continuous dividends at a rate q. Modify the plot section so that it now plots the graph of a call with the same parameters as before but with the dividend yield q=3.2% and the new strike price 11.50. Submit printouts of code and graph.

3. Using arbitrage arguments explain why the price of an American call option on a stock paying no dividends should be the same as the price of a corresponding European call. Why American calls on a nondividend paying stock should not be exercised early.

4. Why when the stock pays dividends the argument of the problem No.3 can not be used. Give a numerical example (choosing x, k, r, $T-t$, σ) in which it is obvious (without any formulas) that American put price on a nondividend paying stock is larger than the corresponding European put price.

5. (a) The stock price is 100 the volatility of the stock is 24%. Assuming that the time to expiration is 3 months and the interest rate is 4.1% per annum calculate the price P of the
European call option with strike 101.

(b) Calculate Δ, Γ, ρ, Vega using formulas for these parameters. Calculate the same parameters approximately using the options calculator.

(c) Check that following relationship holds

$$\Theta + rx\Delta + \frac{1}{2}\sigma^2 x^2 \Gamma = rP$$

6. What are the parameters affecting prices European and American calls and puts. How do the prices change when one of the parameters changes with all the others remaining the same?

7. Suppose that we have three European calls with strikes 30, 35, and 40 and the same maturity 1 month. Their prices are 10.00, 8.00, 5.00. Is it possible to do an arbitrage?

8. Suppose that current stock price is 50 $. Its annualized volatility is 22% and annualized return 11 % i.e. we assume that the stock price follows $dX_t = 0.11\, X_t\, dt + 0.22\, X_t\, dW_t$. Write the probability density function for the stock in 1 year. What is the mean and standard deviation of the terminal stock price? (standard deviation of price, not of return)